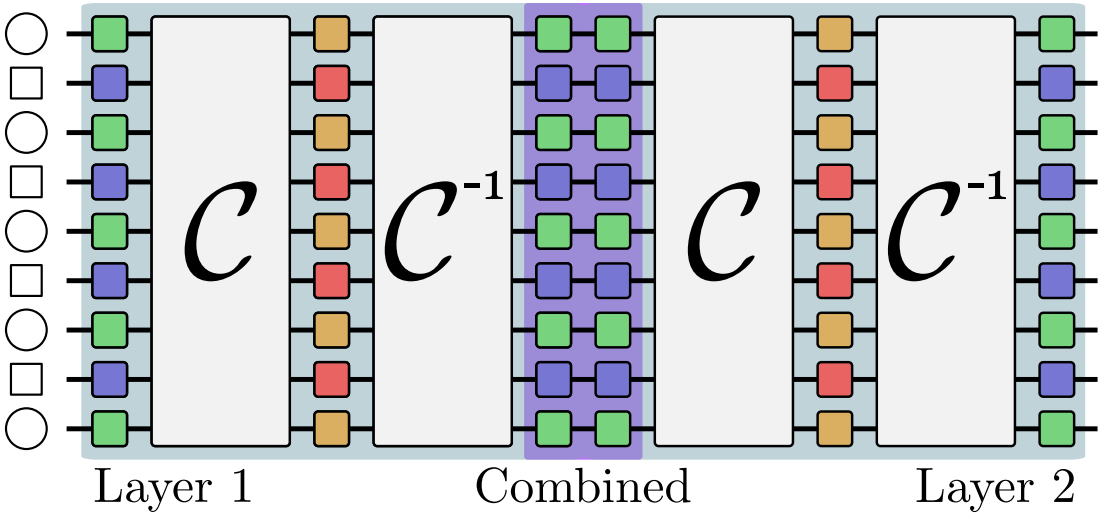

Matter qubits

Gauge qubits

$R_n^z(mt) = \exp(-imZ_m t)$

$R_{(n,n+v)}^z(gt) = \exp\left(-igZ_{(n,n+v)}t\right)$

$R_{(n,n+v)}^x(\lambda t) = \exp\left(-i\lambda X_{(n,n+v)}t\right)$

$R_n^z(\tau_n) = \exp(-iZ_n\tau_n)$